



Porting Driver code to SHAKTI

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Why should I port driver codes?

Factors to Keep in Mind

- **Communication Protocols** – SPI, I2C, UART may require minor changes.
- **Data Type Sizes** – Ensure compatibility between different microprocessor architectures.
- **Endianness** – Verify byte order matches between source and target.
- **Memory-Mapped Registers** – Address mapping may differ.

Steps to Port Driver Code

- **Analyze Existing Driver** – Identify protocol and hardware-specific code.
- **Update Protocol Implementation** – Replace SPI/I2C/UART routines with Shakti equivalents.
- **Adjust Data Types** – Match int, long, char sizes to Shakti.
- **Test & Validate** – Verify functionality on Shakti hardware.

Adapting Protocols for Shakti (Contd)

SPI Example:

```
void write_command(uint8_t value)
{
    // printf("\n [DEBUG] Write Command \n");
    turn_off_gpios(GPIO2);
    // sspi_configure_tx_setup_time(sspi_instance[SSPI_INSTANCE], 2);
    // sspi_configure_tx_hold_time(sspi_instance[SSPI_INSTANCE], 2);
    sspi_configure_tx_rx_length(sspi_instance[SSPI_INSTANCE], 8, 0); //Tx - 8, Rx - 0;
    sspi_configure_comm_mode(sspi_instance[SSPI_INSTANCE], SIMPLEX_TX);
    sspi_instance[SSPI_INSTANCE]->data_tx = value; //MSB byte of address.
    sspi_enable_txx(sspi_instance[SSPI_INSTANCE], ENABLE);
    sspi_notbusy();
    return 1;
}

void write_data8(uint8_t value)
{
    // printf("\n [DEBUG] Write Data \n");
    turn_on_gpios(GPIO2);
    // sspi_configure_tx_setup_time(sspi_instance[SSPI_INSTANCE], 2);
    // sspi_configure_tx_hold_time(sspi_instance[SSPI_INSTANCE], 2);
    sspi_configure_tx_rx_length(sspi_instance[SSPI_INSTANCE], 8, 0); //Tx - 8, Rx - 0;
    sspi_configure_comm_mode(sspi_instance[SSPI_INSTANCE], SIMPLEX_TX);
    sspi_instance[SSPI_INSTANCE]->data_tx = value; //MSB byte of address.
    sspi_enable_txx(sspi_instance[SSPI_INSTANCE], ENABLE);
    sspi_notbusy();
    return 1;
}

void write_command_with_data(uint8_t cmd, const uint8_t *data, size_t len) {
    write_command(cmd);
    for (size_t i = 0; i < len; ++i) {
        write_data8(data[i]);
    }
}
```

Handling Data Types

- Check sizes of int, long, float
- Use stdint.h types (uint8_t, int16_t) for portability
- Example:

Original: int16_t sensor_value;

- **Shakti:** same, but ensure register read/write aligns with 16-bit width

Ensuring Correct Functionality

- Unit test functions individually
- Integrate driver and test sensor communication
- Use logging or debug interfaces on Shakti
- Capturing and analyzing waveform using Logic Analyzer

Key Takeaways

- Porting drivers involves changing only the protocol interface and type handling
- Validate thoroughly on Shakti hardware
- Benefits:
 - code reuse
 - reduced errors
 - faster development

Worked Demo



Thank you

Website: shakti.org.in

GitLab: gitlab.com/shaktiproject